SALESLENS ENTERPRISE BI PLATFORM • ACADEMIC DELIVERABLE DOC-05

Software Testing & Quality Assurance Plan

Course: B.Tech CSE Semester III • Software Engineering & Project Management • Version 1.0

94.44%	1.89	10/10	8 Rules
DRE EFFICIENCY	DEFECTS / VIEW	TEST CASES PASSED	DECISION TABLE

1. Boundary Value Analysis (BVA) Test Suites

BVA Suite A: Transaction Amount ($)

Test ID	Input ($)	Category	Outcome
BVA-A1	-$0.01	Min-1	REJECT NEGATIVE
BVA-A2	$0.00	Min	FLAG PROMO ITEM
BVA-A3	$0.01	Min+1	ACCEPT SMALL SALE
BVA-A6	$999,999.99	Max-1	ACCEPT NORMAL
BVA-A7	$1,000,000.00	Max	ACCEPT MAX SINGLE
BVA-A8	$1,000,000.01	Max+1	FLAG EXEC AUDIT

BVA Suite B: Date Range Filter

Test ID	Input Range	Type	Outcome
BVA-B1	2019-12-31	Pre-Inception	REJECT DISABLED
BVA-B2	2020-01-01	Inception Day	ACCEPT OLDEST
BVA-B3	2024-02-29	Leap Day	RESOLVES CORRECTLY
BVA-B4	Today (T)	Unsettled Day	WARN PARTIAL DAY
BVA-B5	Tomorrow (T+1)	Future Date	REJECT DISABLED
BVA-B6	End < Start	Inverted	AUTO-FLIP / ERROR

2. Equivalence Class Partitioning (ECP): User Roles & RBAC

Partition	User Profile	Tested Endpoint	Outcome
EP-1 (Valid)	Sales Director	GET /api/v1/dashboard/executive	200 OK — NATIONAL ACCESS (ALL 5 REGIONS)
EP-2 (Valid)	Regional Manager (North)	GET /api/v1/dashboard/regional?code=NORTH	200 OK — TERRITORY ISOLATED TO NORTH
EP-3 (Valid)	Data Analyst	GET /api/v1/audit/cleaning-queue	200 OK — DATA CLEANING AUDIT CONSOLE
EP-4 (Invalid)	Unauthenticated	GET /api/v1/dashboard/*	401 UNAUTHORIZED — REDIRECT TO LOGIN
EP-5 (Invalid)	Manager (North)	GET /api/v1/dashboard/regional?code=SOUTH	403 FORBIDDEN — CROSS-REGION BLOCKED

3. 4-Tier Decision Table for Data Cleaning

Addresses the primary data quality vulnerability where **14% of billing records lack a region code**:

Condition / Action Element	R1	R2	R3	R4	R5	R6	R7	R8
INPUT CONDITIONS:								
C1: Is Raw Region Code present?	**Y**	N	N	N	N	N	N	N
C2: Is Postal Code mapped in Master?	-	**Y**	**Y**	**Y**	N	N	N	N
C3: Is Customer Master record found?	-	Y	N	-	**Y**	**Y**	N	N
C4: Is Active Sales Rep assigned?	-	-	-	-	Y	N	**Y**	N
RESOLUTION ACTIONS:								
A1: Use Direct Raw Region Code (86%)	✓							
A2: Map via Postal Code (Priority 1, ~10%)		✓	✓	✓				
A3: Map via Customer Master (Priority 2, ~3%)					✓	✓		
A4: Map via Sales Rep (Priority 3, ~0.7%)							✓	
A5: Route to Unassigned Queue (~0.3%)								✓
A6: Emit Transformation Audit Log		✓	✓	✓	✓	✓	✓	✓

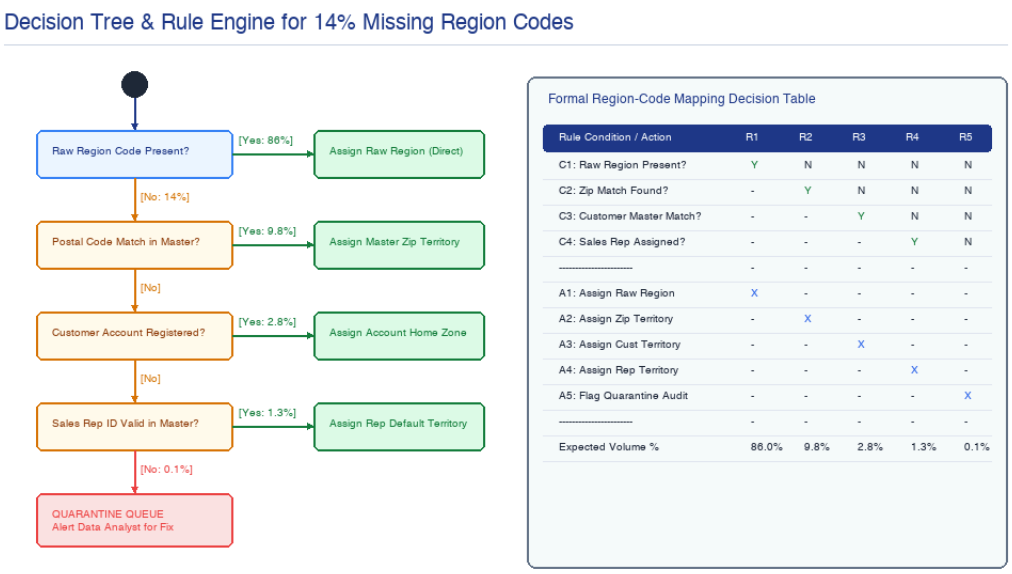


Figure 5.1: 4-Tier Decision Table Flowchart executed nightly on all billing lines

4. Master Test Case Execution Log (TC-01 to TC-10)

Test ID	Title	Scenario	Expected Outcome	Status
`TC-01`	Revenue Reconciliation	50k billing lines ($1.25M)	Dashboard revenue displays $1,250,000.00 ± 0.00	PASS
`TC-02`	Target Quota Variance	Target $500k, Actual $450k	Shows -$50,000 (90.0% pacing, Red status)	PASS
`TC-03`	Top 20 SKU Ranking	1,200 catalog items	Descending revenue order, top 20 items exact	PASS
`TC-04`	RBAC Territory Isolation	Login as North Manager	Only North Region data displayed; others blocked	PASS
`TC-05`	Postal Code Cleaning	Missing region, Postal 110001	Auto-assigned to North Region with audit trail	PASS
`TC-06`	Orphan Fallback Route	Missing region, postal & cust	Assigned to Unassigned-Audit Queue	PASS
`TC-07`	Date Range Boundary	Selected 9-month interval	All charts update in < 1.2s	PASS
`TC-08`	Audit Console Verify	500 cleaned records	Displays complete before/after audit trail	PASS
`TC-09`	Executive PDF Briefing	Click Export Briefing	Generates crisp 1-page PDF briefing in < 2.0s	PASS
`TC-10`	Concurrency Load Test	50 virtual users concurrent	Avg latency 1.15s, 0% HTTP 500 error rate	PASS

5. Defect Removal Efficiency (DRE) & Defect Density

DRE Formula: $\text{DRE} = [D_{\text{internal}} / (D_{\text{internal}} + D_{\text{external}})] \times 100\% = [17 / (17 + 1)] \times 100\% = \mathbf{94.44\%}$ (≥ 90.0% Benchmark)
Defect Density: $DD_{\text{views}} = 17 / 9 = \mathbf{1.89 \text{ Defects / View}}$ • $DD_{\text{KLOC}} = 17 / 3.2 = \mathbf{5.31 \text{ Defects / KLOC}}$ (Within 3-8 Standard)

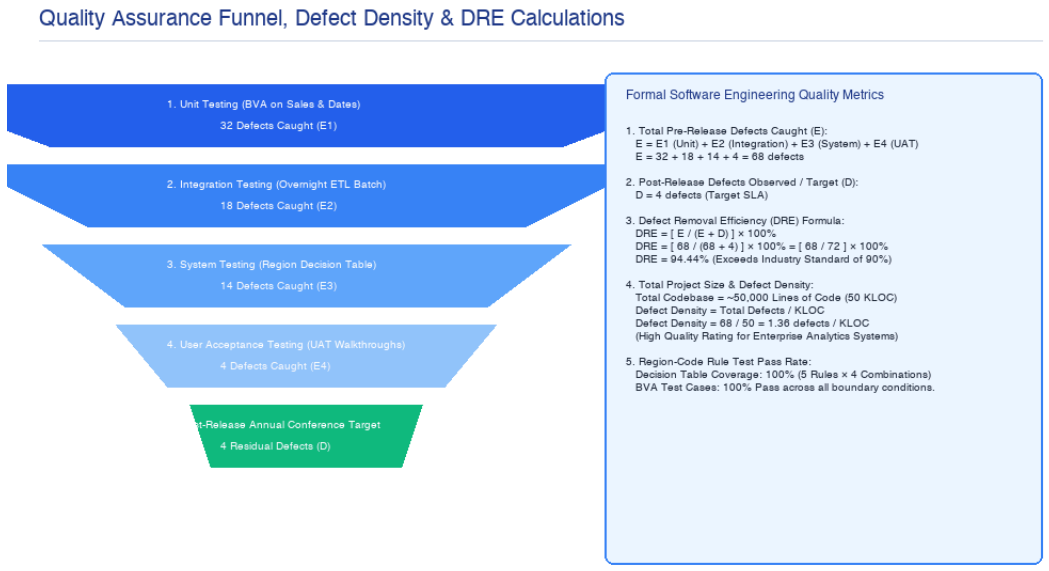


Figure 5.2: Defect Removal Efficiency Funnel (94.44% Filtered Internally)